

ARLINGTON, TX, USA

WMO: 722479

Lat: 32.664N

Lon: 97.094W

Elev: 630

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53907

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.8	27.6	8.9	8.9	33.8	12.5	10.6	38.1	26.5	49.2	24.4	52.3	7.9	340	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.7	101.1	73.8	99.1	74.2	96.9	74.2	78.1	90.8	77.3	90.2	76.6	89.2	9.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	134.1	82.0	73.9	129.3	81.5	73.2	126.1	81.4	41.8	91.0	41.0	90.7	40.3	89.2	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.9	20.7	18.8	DB	17.0	104.0	3.9	3.5	14.2	106.5	11.9	108.6	9.8	110.5	7.0	113.1
			WB	15.2	79.6	3.4	1.1	12.7	80.4	10.7	81.0	8.8	81.7	6.2	82.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.9	47.0	50.6	58.7	66.4	74.0	81.7	84.9	85.7	78.7	67.8	57.3	48.5
	DBStd	15.70	9.77	10.74	9.22	7.38	6.11	4.06	3.71	4.45	6.06	8.34	9.43	9.43
	HDD50	497	172	112	29	2	0	0	0	0	0	2	37	143
	HDD65	2133	560	413	234	70	10	0	0	0	1	68	261	516
	CDD50	6656	80	130	300	495	745	950	1081	1106	861	554	257	97
	CDD65	2817	3	11	40	112	291	500	616	641	412	155	30	6
	CDH74	32880	31	79	278	864	2684	5851	8143	8683	4590	1439	201	37
	CDH80	16390	2	11	44	200	966	2881	4535	4999	2220	496	32	4
(14) Wind	WSAvg	9.1	9.2	10.0	10.5	11.0	10.3	9.3	8.1	7.4	7.3	8.2	8.8	8.9
(15) Precipitation	PrecAvg	37.8	2.3	2.5	3.1	3.7	4.3	3.9	2.3	2.3	3.1	4.0	2.9	2.8
	PrecMax	55.1	4.6	6.9	5.9	7.4	8.3	11.1	6.1	5.8	10.4	9.8	7.3	9.5
	PrecMin	19.0	0.2	0.0	0.0	1.4	0.4	0.7	0.0	0.0	0.2	0.0	0.5	0.2
	PrecStd	9.2	1.3	1.9	1.7	1.7	2.0	3.2	1.7	1.4	2.8	2.8	2.1	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.0	81.6	84.2	89.2	94.1	99.4	103.5	105.0	100.8	92.9	83.9	78.5
		MCWB	58.9	60.7	64.4	69.6	73.6	74.1	73.7	72.6	72.3	71.1	66.1	63.7
	2%	DB	72.8	76.1	80.8	84.7	90.8	96.7	100.4	101.9	96.8	89.4	79.3	73.1
		MCWB	58.2	60.9	64.2	67.9	72.8	74.2	74.1	73.7	72.8	71.0	64.6	61.1
	5%	DB	68.7	72.2	77.1	82.0	88.2	94.4	98.4	99.6	93.6	85.4	75.4	69.0
		MCWB	57.5	59.7	62.7	67.4	72.2	74.1	74.6	74.3	72.3	68.5	63.2	59.0
	10%	DB	64.2	68.2	73.4	79.2	85.5	92.2	95.9	97.4	91.0	82.0	72.3	64.6
		MCWB	54.8	59.0	61.5	65.9	71.2	74.1	74.7	74.2	72.1	67.2	62.6	56.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.5	67.9	69.4	73.9	77.4	79.3	79.0	78.8	77.9	75.5	70.7	67.4
		MCDB	71.4	74.4	78.7	83.5	88.8	91.6	91.2	92.2	88.6	84.3	76.6	73.2
	2%	WB	63.7	65.8	67.5	71.8	75.6	77.7	77.9	77.6	76.5	74.1	68.3	65.0
		MCDB	68.6	71.5	75.7	80.8	85.5	89.7	91.1	91.6	87.1	82.8	74.5	70.7
	5%	WB	60.2	63.5	66.1	70.1	74.5	76.7	77.2	76.9	75.5	72.6	66.4	61.7
		MCDB	66.0	69.1	73.5	78.1	84.2	88.4	90.5	90.8	85.9	81.2	73.1	66.3
	10%	WB	55.2	60.3	64.2	68.6	73.3	75.8	76.4	76.2	74.6	70.7	64.2	57.7
		MCDB	62.9	66.6	71.4	76.1	82.7	87.0	89.4	89.8	85.0	78.8	70.7	63.8
(35) Mean Daily Temperature Range	5% DB	MDBR	21.2	20.5	20.7	20.7	19.1	18.7	20.3	20.7	20.7	21.7	21.0	20.3
		MCDBR	27.9	25.5	23.8	22.7	21.4	21.4	23.3	23.6	23.0	24.1	23.5	25.7
		MCWBR	16.7	15.2	12.2	11.1	8.5	5.8	5.7	5.8	6.6	10.2	12.4	15.9
	5% WB	MCDBR	22.8	21.4	19.6	17.7	18.2	17.7	19.9	20.1	18.7	18.9	19.0	19.8
		MCWBR	17.1	15.6	12.5	10.7	8.6	6.1	5.9	5.7	6.5	9.1	13.0	16.2
(40) Clear-Sky Solar Irradiance	taub		0.314	0.328	0.357	0.395	0.425	0.436	0.439	0.439	0.400	0.359	0.338	0.319
	taud		2.502	2.470	2.406	2.323	2.299	2.301	2.331	2.314	2.371	2.472	2.472	2.489
	Ebn at Noon		288	294	292	284	275	270	269	268	274	279	276	279
	Edh at Noon		27	31	36	41	42	42	40	41	37	31	28	26
(44) All-Sky Solar Radiation	RadAvg		921	1077	1422	1724	1858	2090	2132	1955	1642	1290	974	811
	RadStd		91	171	111	136	173	196	145	127	138	192	124	105

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page